HySens: Sensor-Agnostic Foundation Models for Hyperspectral Data

Xiangyu Zhao, Zhitong Xiong, *Member, IEEE*, and Xiao Xiang Zhu, *Fellow, IEEE*

***Abstract*—Foundation models (FMs) have been extensively explored and have demonstrated great performance in various remote sensing tasks. Among all the types of remote sensing data, there are relatively few studies on hyperspectral FMs. One big challenge is that different hyperspectral sensors have significantly different sensor characteristics in both spectral and spatial domains. Such discrepancies make it difficult to develop a universally applicable hyperspectral FM, which could be easily transferred and applied to datasets from diverse hyperspectral sensors. In this work, we develop HySens: a generic sensor-agnostic hyperspectral FM that can be applied easily and efficiently to various hyperspectral sensors. Specifically, 1) we integrate a spectral harmonizer by harmonizing various bands distribution into a unified spectral grid with high fidelity and less redundancy; 2) we design customized wavelength-based spectral embeddings such that the spectral embeddings are wavelength aware and could be easily generated based on the physical bands wavelengths; and 3) we treat the hyperspectral spectra as sequences and focus on the spectral sequential features by using a transformer backbone. The proposed FM is compared with other representative hyperspectral algorithms and recent hyperspectral FMs over both patch-based and pixel-based hyperspectral tasks. The developed model achieves superior performance with good interpretability and efficiency.**

***Index Terms*—Foundation model (FM), hyperspectral, sensor agnostic, spectroscopy.**

I. Introduction

HYPERSPECTRAL data are currently used for many applications, including crop type mapping, soil property estimation, and water property detection [1]. The success of such applications could be attributed to the fine spectral details stored in the hyperspectral datasets. Unlike many other remote sensing modalities, such as optical images, multispectral images, or synthetic aperture radar (SAR), hyperspectral datasets always contain several hundred spectral bands. Such abundant spectral information enables the extraction of subtle spectral features that are hardly detectable with other data modalities [2]. The unique spectral properties of hyperspectral data make them highly useful for a wide range of applications.

However, on the other side, hyperspectral datasets are limited, sparsely distributed, and highly heterogeneous. Hyperspectral data collection is always task-oriented, focusing on targeted regions rather than continuous global coverage. For now, public hyperspectral datasets can be broadly categorized into two types. The first are large-scale spaceborne datasets, such as data from EnMAP [3], DESIS [4], and PRISMA [5] satellites, which are sparsely distributed globally and unlabeled. The second are many small airborne datasets, which always contain only a few scenes and are collected for specific applications with task-specific labels [6], [7]. Such data scarcity and inconsistency have led to the development of many sensor- and task-specific hyperspectral models, which are hardly transferable and not generally applicable. Also, due to such data discrepancies and limitations, there are many studies working on cross-domain hyperspectral classifications [8], [9], [10], cross-domain fusion [11], and domain generalizations [12]. These data situations and challenges are less pronounced in other remote sensing modalities, such as multispectral datasets from Sentinel-2 [13] or Landsat [14], where they always have continuous and global data collections.

Deep learning methods have demonstrated great performance in many vision and language applications. Based on the success of vision methods, including convolutional neural networks (CNNs) [15], vision transformers [16], etc, many deep learning frameworks have been successfully adapted for remote sensing data. More recently, foundation models (FMs) have gained popularity across multiple domains, where large-scale unlabeled datasets are leveraged to pretrain high-capacity models. These pretrained models can then be efficiently transferred to diverse downstream tasks. However, it becomes uniquely challenging when developing deep learning models for hyperspectral data [17]. First, hyperspectral imagery differs substantially from conventional optical imagery, with hundreds of spectral channels compared to only three in RGB images. Consequently, the key information in hyperspectral data is primarily stored in the spectral domain, whereas vision-based methods emphasize the spatial features. Many successful vision frameworks, therefore, require significant and careful adaptation to effectively process hyperspectral data. Second, labeled hyperspectral datasets are typically very limited in size. For example, hyperspectral classification datasets often consist of a single image with pixel-wise annotations [6], [7], making it challenging to train large deep learning models with

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numerous trainable parameters. These challenges highlight the need and potential of building sensor-agnostic hyperspectral FMs. By pretraining on large-scale unlabeled spaceborne hyperspectral data, the FMs could learn general spectral representations that transfer effectively to various small-sized datasets for specific applications. Moreover, the sensor-agnostic property enables applicability across hyperspectral sensors with different characteristics, greatly enhancing the FM's practical utility for end users with diverse hyperspectral applications.

To address the challenges of multimodality or multisensor data in remote sensing, many NEWmethods have been developed [18] in existing studies. Some are multimodal NEWmodels, incorporating optical, multispectral, or even SAR imagery as inputs. Works such as SkySense [19], TerraMind [20], NEWAlliance [21], STARS [22], and Copernicus-FM [23] NEWemploy Vision-based backbones like ViT and demonstrate strong performance across various data modalities. However, these models typically do not include hyperspectral data, and their pretrained representations are primarily spatial rather than spectral, which makes them less aligned with the characteristics of hyperspectral data and the needs of hyperspectral applications. There are some other FMs focusing on hyperspectral data, such as HyperSigma [24], SpectralEarth [25], and HyperSL [26]. Although these models could be applied directly across different hyperspectral sensors, their architectures typically rely on simply converting the original hyperspectral inputs into fixed-size feature representations, without explicitly accounting for differences in spectral domains across sensors. Such a straightforward transformation could be sufficient for relatively easy classification tasks, but it could limit transfer learning performance in more challenging spectroscopy applications, where detailed spectral information is necessary, and important [2].

Building on the discussions above, several key research questions arise: 1) When the pretrained and finetuned hyperspectral data are highly different, how can the pretrained model be transferred effectively? 2) Beyond simply transforming hyperspectral data into fixed-sized features, are there any flexible ways to incorporate the sensor-specific differences and generate features dynamically? 3) Given the challenging spectroscopy tasks, which frameworks are suitable for hyperspectral FMs? and 4) If the finetuning tasks contain hyperspectral patches, how can the spatial information be integrated effectively without losing the spectral fidelity? Regarding these questions and challenges, this study aims to develop a sensor-agnostic FM for hyperspectral data, which can be applied effectively and efficiently across various hyperspectral sensors and diverse hyperspectral applications. Our main contributions are as follows.

1) We introduce a spectral harmonizer which aligns heterogeneous band distributions onto a unified spectral grid, ensuring high fidelity and reduced redundancy.
2) We design customized wavelength-based spectral embeddings, enabling the generation of wavelength-aware representations based on the physical wavelengths across sensors. The positional embeddings are also adapted to explicitly encode the actual physical wavelength information.
3) We treat hyperspectral spectra as 1-D sequences and focus on extracting spectral sequential features using a transformer-based spectral encoder.

Based on such designs, the developed FM is then compared with other state-of-the-art hyperspectral algorithms and FMs on both patch-based and pixel-based tasks, demonstrating superior performance, good interpretability, and improved efficiency. The remainder of this article is organized as follows. In Section I, related works in FMs, sensor-agnostic models in remote sensing, hyperspectral algorithms, and hyperspectral applications are introduced. In Section III, the model design of the proposed FM is explained. In Section IV, the experimental results of using the proposed FM and other baseline methods are shown and compared. Ablation studies about wavelength-based modules, models' explainability, and efficiency are described. In the end, conclusions are drawn up in Section V.

II. Related Work

A. FMs in Remote Sensing

The FM is a high-capacity model pretrained on large-scale unlabeled datasets, with the resulting representations transferable to diverse downstream applications. FMs are initially developed in the fields of natural language processing and computer vision. They have recently gained traction in remote sensing, achieving superior performance in various applications [27]. A key development trend in FM for remote sensing is the integration of multiple data modalities within a unified framework [28]. For example, SkySense [19] has incorporated heterogeneous data, including optical, multispectral, and SAR imagery. By integrating multiple data modalities, pretrained spatial representations can be transferred to downstream tasks such as land cover classification, semantic segmentation, and object detection. TerraMind [20] is a generative, large-scale multimodal FM that incorporates nine geospatial modalities, achieving superior performance on benchmark datasets such as PANGAEA [29]. Besides these models, many other works, such as RingMo [30], RemoteClip [31], UniTS [32], and DOFA [33] are all multimodal FMs. Regarding different data modalities, these models commonly project each modality into a shared feature space, enabling effective alignment and joint learning across multiple data sources. Such feature representation has demonstrated excellent performance in aligning data from optical data, multispectral imagery, SAR data, and even natural languages. However, hyperspectral data remains highly underexplored in these FMs due to their distinctive data characteristics and the limited availability of large-scale hyperspectral datasets.

B. Sensor-Agnostic Models in Remote Sensing

Compared with other research fields, remote sensing data often come from multiple sensors with widely varying spectral and spatial characteristics. To address these challenges, several works have explored sensor-agnostic models. One common approach is to explicitly encode sensor parameters into the feature space. For instance, Senpa-MAE [34]

encodes sensor parameters of multispectral images into image embeddings. Specifically, the spectral response function and the ground sampling distance are considered when generating sensor-independent embeddings. Similarly, STARS [35] has developed a sensor-agnostic transformer by introducing a universal spectral representation derived from sensor metadata. While these strategies can effectively leverage sensor parameters, the additional model complexity can reduce efficiency, especially when applying to small datasets. Another sensor-agnostic approach is to design customized tokenizers for different input data. For example, SMARTIES [36] is a generic and versatile FM that projects data from heterogeneous sensors (RGB, multispectral, SAR) into a shared spectrum-aware space. HyperSL [26] proposes a universal spectral tokenizer to transform spectral vectors of varying dimensions into a standardized token format, enabling a single model to process multisource hyperspectral data. These tokenizer-based approaches offer dynamic methods for generating embeddings from diverse spectral data. However, such designs remain largely unexplored for hyperspectral FMs, and existing adaptations often lack wavelength-aware modules tailored to the hundreds of contiguous bands in hyperspectral data.

C. Hyperspectral Algorithms and Hyperspectral Applications

In current deep learning studies of hyperspectral data, there are generally two approaches: pixel-based and patch-based. In pixel-based methods, the input is a 1-D sequence representing the hyperspectral spectrum of a single pixel. Sequential models, such as RNN [37] and transformer [38], have been successfully adapted to extract spectral features for hyperspectral classification. In patch-based methods, the input is a 3-D hyperspectral cube. By compressing the spectral bands into a few channels, the hyperspectral patch can be transformed to resemble a natural image, enabling the application of vision-based methods. For example, Chen et al. [39] proposed a deep belief network to extract spectral–spatial features, while Peng et al. [40] introduced a spatial–spectral cross attention module with a dual-branch structure. Most current hyperspectral algorithms are evaluated on land cover classification benchmarks [41], where the datasets consist of only one or a few annotated hyperspectral images. While such evaluations are sufficient for comparing hyperspectral methods, they do not fully reflect the complexities of real-world hyperspectral applications. Spectroscopy-based tasks are an important hyperspectral application in which ground materials are identified by their spectral signatures. By using spectra from hyperspectral data, properties such as soil composition [42], vegetation indices [43], [44], or water quality [45] can be estimated. In many of these scenarios, the datasets are typically small, the annotations are spatially sparse, and the discriminative features lie primarily in the spectral domain, with spatial information available only in certain cases. Such data conditions and task requirements differ sharply from the commonly used hyperspectral classification datasets, where dense pixel-wise labels and strong spatial correlations make patch-based methods effective. Therefore, real-world hyperspectral applications impose different algorithmic requirements: models should be lightweight, prioritize effective spectral feature extraction, and be able to leverage available spatial information.

III. Methodology

In this section, the methodology design is explained in detail in correspondence to the research questions we raised in Section I.

1) To make transfer learning effective across various hyperspectral datasets, a spectral harmonizer is first introduced (Section III-A).
2) To incorporate sensor-specific differences, wavelength-based spectral embedding layers (Section III-B) are designed to dynamically generate spectral representations. The decoder follows a symmetric structure design to help preserve sensor-specific spectral sequential information (Section III-D).
3) Given the data conditions and task requirements of spectroscopy applications, the transformer framework serves as the backbone for the spectral encoder (Section III-C).
4) When the hyperspectral patches are available for downstream applications, a spectral spatial aggregator (Section III-E) would be deployed to spatially fuse spectral features.

Following the answers of the proposed research questions, the overall framework of the proposed sensor-agnostic hyperspectral FM is illustrated in Fig. 1 for pretraining and Fig. 2 for finetuning. In general, the model is designed primarily to capture spectral features, which store highly relevant information for many hyperspectral applications. To address differences among hyperspectral sensors, several dedicated modules are introduced. As shown in Fig. 1, first, a spectral harmonizer aligns the distributions of different spectral bands onto a unified spectral grid. Second, a group masking strategy divides all spectral bands into groups, which are randomly masked during pretraining. Third, the wavelength-based embedding layers project group reflectance values into the feature space in a wavelength-aware manner, complemented by customized wavelength-based positional embeddings to reflect the physical sequential information. Fourth, a spectral encoder is integrated to enrich the wavelength-based features with spectral context information. Ultimately, the wavelength-based decoder layers reconstruct the spectrum from the feature space during pretraining, following a symmetric structure with the embedding layers. As shown in Fig. 2, during fine-tuning, the pretrained modules are transferred and applied to both patch-based and pixel-based tasks. In particular, a spectral spatial aggregator would be employed to generate compact, representative spectral spatial features for patch-based hyperspectral applications.

A. Spectral Harmonizer

For hyperspectral sensors, the spectral resolutions typically range from around 1–5 nm in airborne sensors to around 5–10 nm in spaceborne sensors. Even within a single sensor, the resolutions often differ between the visible and near-infrared (VNIR) and short-wave infrared (SWIR) ranges. Given such spectral resolution differences, the spectral features

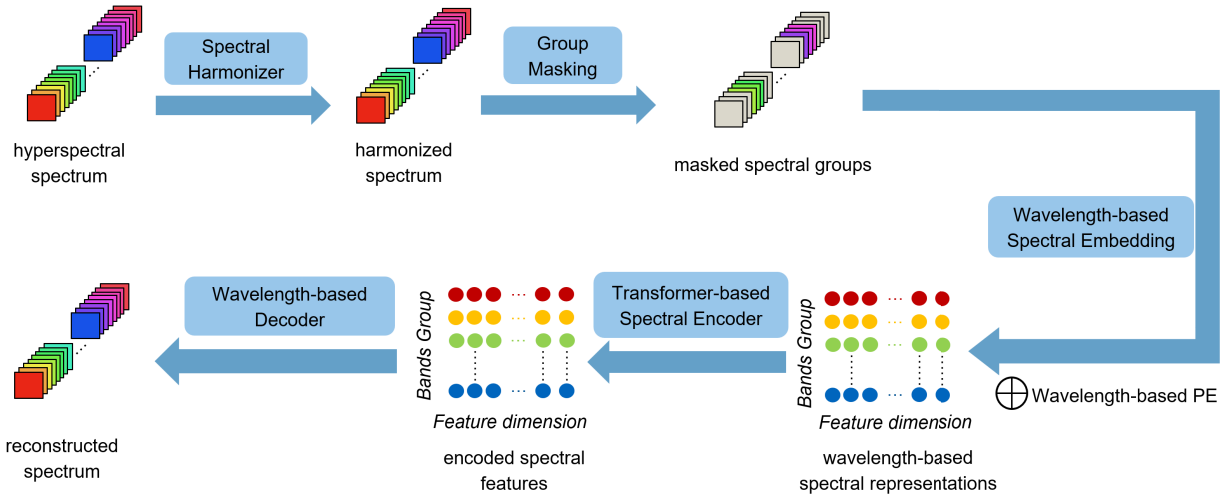


Fig. 1. Pretraining pipeline for the sensor-agnostic hyperspectral FM: several modules are designed to cope with the hyperspectral sensor differences, such as spectral harmonizer, wavelength-based spectral embedding, and wavelength-based decoder. The entire framework is pretrained using masked spectral modeling, with a focus on spectral feature extraction.

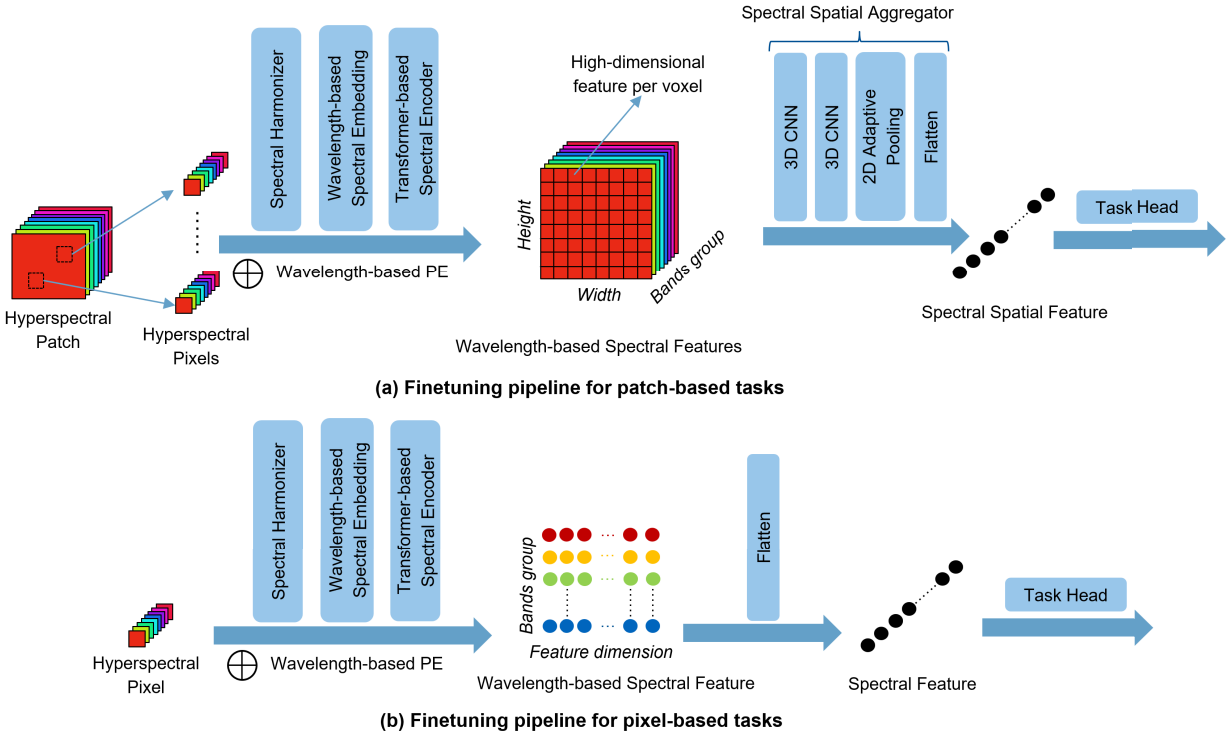


Fig. 2. Finetuning pipelines. (a) Finetuning pipelines on patch-based hyperspectral applications, to make use of the spatial information, a spectral spatial aggregator is employed before the task head to generate the spectral spatial feature. (b) Finetuning pipelines on pixel-based hyperspectral applications, the pixel-based spectral features are generated using the pretrained FM, and a simple task head is attached.

extracted by a pretrained model and those required by downstream applications would be on different scales. Such scale mismatches would make the transfer learning less effective when applying a pretrained FM across diverse sensors. Therefore, to ease the difficulty of transfer learning across sensors, the spectral resolutions, as well as the level of spectral details, should be carefully aligned. On the one hand, the original spectra of hyperspectral data are often redundant for relatively easy tasks like land cover classification [17]. The spectra could then be compressed without losing essential information. On the other hand, for demanding spectroscopy tasks, such as estimating soil properties [46] or water properties [47], preserving spectral fidelity is necessary to retain task-relevant spectral features.

Considering the wide range of spectral resolutions and diverse task requirements, we design a spectral harmonizer that aligns all input spectra into a unified 10-nm grid by interpolating missing bands and removing redundant ones. Based on the spectral characteristics of commonly used hyperspectral sensors—ranging from airborne to spaceborne platforms and from VNIR to SWIR regions [48], the 10-nm grid represents a practical and robust harmonization choice across

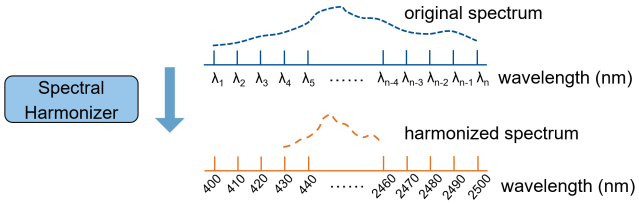


Fig. 3. Spectral harmonizer. Based on the spectral bands distribution of different sensors, the spectral harmonizer would align various input spectra into harmonized spectra with a 10-nm grid in the range of 400–2500 nm.

heterogeneous sensors. Many airborne hyperspectral sensors with finer spectral sampling exhibit strong interband redundancy [49] due to overlapping spectral response functions. In contrast, spaceborne hyperspectral sensors generally have coarser spectral resolutions and contain missing or irregularly distributed bands due to sensor and atmospheric constraints [50]. The proposed harmonization alleviates these inconsistencies by interpolating missing bands without artificial oversampling. The choice of 10-nm resolution can provide a balanced tradeoff between spectral redundancy reduction and spectral fidelity across sensors of various band configurations. Since most hyperspectral sensors are within the spectral range from 400 to 2500 nm, the harmonized grid is established within this interval. The whole process is described by the following mathematical formulas and visualized in Fig. 3.

Let an input spectrum be represented as

$$\mathbf{x} = \{(\lambda_c, r_c)\}_{c=1}^{C}$$

where $\lambda_c \in [400, 2500]$ denotes the center wavelength of the cth band and r_c is the corresponding reflectance value. The spectral sampling $\{\lambda_c\}$ varies across sensors.

We define a unified spectral grid

$$\hat{\Lambda} = \{\hat{\lambda}_k = 400 + 10(k-1)\}_{k=1}^{K}$$

covering the wavelength range [400, 2500] nm.

The harmonized spectrum

$$\hat{\mathbf{x}} = \{\hat{r}_k\}_{k=1}^{K}$$

is obtained by mapping $\mathbf{x}$ onto $\hat{\Lambda}$ as

$$\hat{r}_k = \begin{cases} r_{c^*}, & |\hat{\lambda}_k - \lambda_{c^*}| < 10\,\text{nm} \\ r_i + \dfrac{\hat{\lambda}_k - \lambda_i}{\lambda_j - \lambda_i}(r_j - r_i), & \text{otherwise} \end{cases}$$

where $c^* = \arg\min_c |\lambda_c - \hat{\lambda}_k|$, and λ_i and λ_j are the original wavelengths immediately below and above $\hat{\lambda}_k$.

Specifically, when the input spectra have a higher spectral resolution (bandwidth < 10 nm), redundant bands within each 10-nm interval are removed to compress the spectral information. Conversely, when the input spectra have a coarser spectral resolution (bandwidth > 10 nm) or missing bands, reflectance values are generated at the missing wavelengths through linear interpolation. It is noted that the spectral harmonizer is designed as a deterministic spectral alignment module rather than an optimized spectral interpolation method. Harmonization is based on linear interpolation due to its numerical stability and common use in hyperspectral processing. Through this harmonization process, the bands of all spectra are mapped onto a unified spectral reference grid, harmonizing feature scales across sensors and reducing domain shifts. This facilitates more effective transfer learning across various hyperspectral sensors.

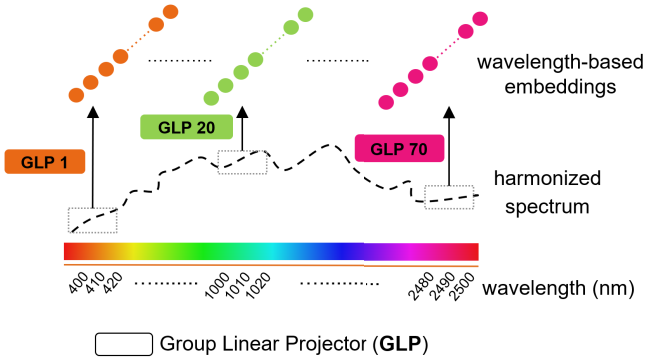


Fig. 4. Wavelength-based spectral embeddings: every three consecutive reflectance values are projected into high-dimensional features using GLP. The generating embeddings are then explicitly aware of the available wavelengths, while keeping the sequential information of the original spectrum.

B. Wavelength-Based Spectral Embedding

In many hyperspectral studies, transformer-based models have been adapted to extract hyperspectral features [17]. However, prior to feeding the reflectance values into the transformer modules, most approaches treat all spectral bands equally when projecting them into a high-dimensional embedding space. For example, in the works of [38] and [51], a simple linear layer is applied as the embedding layer. Papers such as [26] and [52] employ CNN-based embedding layers where identical convolution kernels slide along the spectral dimension. Although such operations can perform well when training and testing on the same sensor, and may even transfer to similar sensors, they struggle when the band distributions differ significantly. The limitation arises since pretrained embedding weights are tuned to capture sensor-specific spectral features, while downstream tasks might require features aligned with a different spectral configuration. Such a mismatch will become particularly problematic for challenging spectrum-driven tasks that require fine-grained spectral features.

Therefore, in our study, we design wavelength-based spectral embedding that maps hyperspectral reflectance values into a high-dimensional embedding space in a wavelength-aware manner. The design is inspired by the field of natural language processing, where pretrained token embeddings are widely applicable across various language tasks. For example, ELMo [53] and BERT [54] both produce contextualized token embeddings that can be directly used by downstream tasks. In these language models, tokens could be words, subwords, or even characters. They serve as the basic input units, each with a consistent semantic meaning across datasets. In the hyperspectral field, after applying the spectral harmonization described in Section III-A, spectra can also be represented as sequences of tokens. However, unlike in the language field, a wavelength's reflectance value varies and depends on the ground surface and acquisition conditions, meaning that

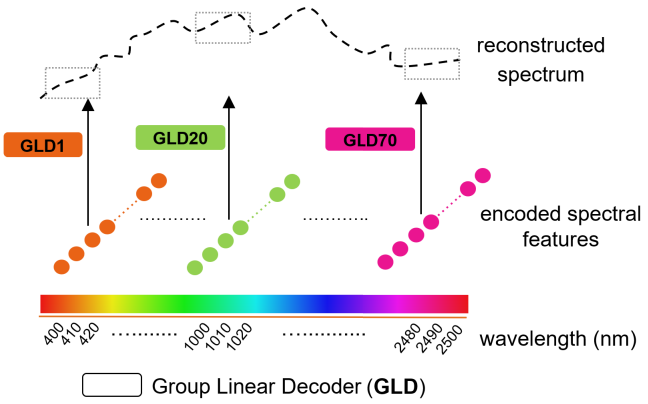


Fig. 5. Wavelength-based decoder: Each encoded spectral feature is projected back to the three consecutive reflectance values using GLD. The decoder follows a symmetric structure with wavelength-based embedding layers.

pretrained embeddings cannot be directly transferred. So here, we make the embedding layers transferable rather than the embeddings. The hyperspectral tokens are embedded in a manner informed by their wavelengths. Specifically, inspired by the group masking strategy on multispectral bands in SatMAE [55], an adapted group masking strategy is deployed here for hyperspectral data. In SatMAE, multispectral bands are grouped by the same spatial resolutions and broad wavelength ranges (RGB NIR, Red Edge, and SWIR), and each spectral band group is treated as an independent image channel. In contrast, our design groups hyperspectral bands based on spectral correlation and continuity, reflecting the dense, sequential nature of hyperspectral spectra. Every three consecutive spectral bands are grouped into a spectral token, which balances fine spectral details preservation and redundant spectral noise suppression. Compared with other group sizes, such a group setting effectively captures local spectral patterns while maintaining computational efficiency. Each spectral token is then projected into a high-dimensional embedding space using a customized group linear projector (GLP), as shown in Fig. 4. Such computation can generate spectral representations that are explicitly aware of the represented wavelengths, while keeping the sequential order of the original spectrum in the embedding space. Such an embedding design would facilitate wavelength-aware representations learning, and the weights of these embedding layers can be easily selected and transferred based on the spectral characteristics of different hyperspectral sensors.

In addition to wavelength-based spectral embedding, positional embeddings are also adapted to explicitly encode the actual physical wavelength information. In transformer-based models, the sequential order is typically stored in position embeddings (PEs). Standard PEs are computed from sequential indices using sine and cosine functions. However, it can be problematic when using such PEs on hyperspectral data. Different hyperspectral sensors have different band distributions. When applying the basic PE computation, a mismatch arises between the physical wavelength positions and the PEs. For instance, when pretraining with spectra in the 400–2500-nm range, the first spectral sequential index corresponds to 400 nm. When fine-tuning with spectra spanning 430–860 nm, the first index corresponds to 430 nm instead. With the standard index-based PE, both hyperspectral datasets share the same PE settings despite representing different wavelengths, leading to mismatches. Therefore, a simple wavelength-based positional embedding is designed to reflect the physically aligned sequential information of hyperspectral data. Specifically, instead of relying on the sequential indices, PEs are computed from the harmonized absolute spectral indices of band wavelengths. Such a PE design can not only provide good continuity and smoothness across wavelengths, but also possess better transfer correspondence across hyperspectral sensors. The computation is described in the following equations, where d is the embedding dimension, k is the dimension index, and λ and i_λ are the wavelength and the corresponding spectral index:

$$i_\lambda = \frac{\lambda - 400}{10} \tag{1}$$

$$\mathrm{PE}_{(i_\lambda, 2k)} = \sin\left(\frac{i_\lambda}{10\,000^{2k/d}}\right) \tag{2}$$

$$\mathrm{PE}_{(i_\lambda, 2k+1)} = \cos\left(\frac{i_\lambda}{10\,000^{2k/d}}\right). \tag{3}$$

This ensures that the same wavelength consistently corresponds to the same PE for any sensors. Such a PE design could reduce sensor mismatches and facilitate effective transfer learning.

C. Transformer-Based Spectral Encoder

After obtaining the wavelength-based features from Section III-B, we incorporate a transformer-based spectral encoder to capture spectral contextual dependencies. The encoder is built on transformer layers, which consist of multihead self-attention and feed-forward modules [56]. Given the wavelength-based features as input, the transformer-based encoder outputs context-aware representations by capturing spectral interactions across the entire spectral range. There are two reasons for using such an encoder structure. First, our FM focuses on spectral features, and transformer-based models have proven highly effective at extracting sequential features from diverse data modalities. Second, compared to many vision-based architectures, the 1-D transformer encoder is structurally simple and parameter-efficient, making it easier to be pretrained and fine-tuned. Such efficiency is particularly important for domain users with limited computational resources.

D. Wavelength-Based Decoder

During pretraining, a wavelength-based decoder reconstructs the masked spectral reflectance from the feature space. To make the decoder wavelength-aware while retaining the spectral sequence of the hyperspectral input, the decoder employs a symmetrical design with wavelength-based embedding layers. Specifically, it consists of multiple group linear decoders (GLDs), as illustrated in Fig. 5. For each encoded spectral group feature, a customized decoder consisting of linear layers projects it back into reflectance values. Such a

design can preserve spectral sequential information and enable wavelength-aware reconstruction of spectral reflectance. It is noteworthy that the decoder is only used during pretraining and not during finetuning.

E. Spectral Spatial Aggregator

For certain hyperspectral applications, such as patch-based land cover classification or change detection, the spatial context provided by pixels surrounding the center target is available and highly informative. Unlike many other vision-based methods that explicitly extract deep spatial features, our model integrates spectral features across the patch's spatial area by utilizing a spectral spatial aggregator. It is only used during the finetuning stage when the surrounding hyperspectral patch is available. It is aimed to aggregate local spatial information to stabilize and enrich the spectral representation. As shown in Fig. 2(a), the spectral spatial aggregator consists of two simple 3-D CNN layers together with a 2-D adaptive pooling layer. The input is the spectral features generated from the spectral encoder. The 3-D CNN layers operate jointly across the spectral dimension (groupwise) and the spatial dimensions (pixelwise), thereby capturing spectral–spatial interactions. The subsequent 2-D adaptive pooling layer further fuses these features across the spatial dimensions to produce a compact and informative spectral–spatial representation for downstream applications. The aggregator is intentionally designed to be simple, in order to maintain an emphasis on spectral information and to ensure efficient finetuning, rather than introducing a dedicated and potentially overspecialized spatial modeling module.

IV. Experiments

In this section, the experimental details are explained, including pretraining of the FM in Section IV, finetuning over different hyperspectral tasks in Section IV-B, and ablation studies that examine the wavelength-based modules and assess the model's explainability and efficiency in Section IV-C.

A. Pretraining

1) Dataset: For pretraining the FM, the HySpecNet-11k dataset [57] is used. This dataset is collected and preprocessed from the EnMAP satellite. It contains 11 483 nonoverlapping hyperspectral patches, each with pixels of 128 × 128, 224 spectral bands, and a ground sampling distance of 30 m. After removing bands affected by strong water vapor absorption, each patch in this dataset contains 202 bands. The reasons for choosing this pretraining dataset are that EnMAP provides continuous and dense spectral coverage across both the VNIR and SWIR ranges, which captures most of the spectral variability encountered in hyperspectral applications. Moreover, using such well-prepared data could provide clean, controlled pretraining settings. Since our approach primarily focuses on the spectral dimension, the pretraining is conducted in a pixel-wise manner. In total, our model is pretrained over 188 million hyperspectral spectra. Before feeding the pixel-based spectral data into the model, the spectral reflectance values are normalized to the range of [−1, 1] and all the spectra are

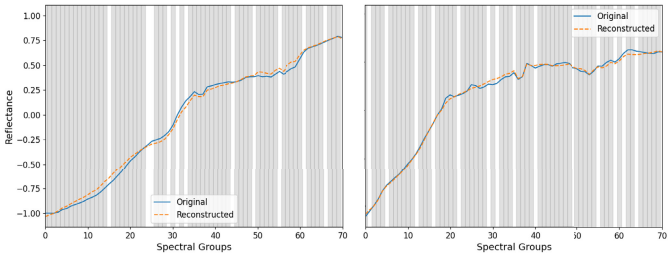


Fig. 6. Original spectra and reconstructed spectra: the reconstructed spectra are generated from the masked spectra using the pretrained FM. The alignment of the overall shape and fine details indicates the effectiveness of the pretrained model.

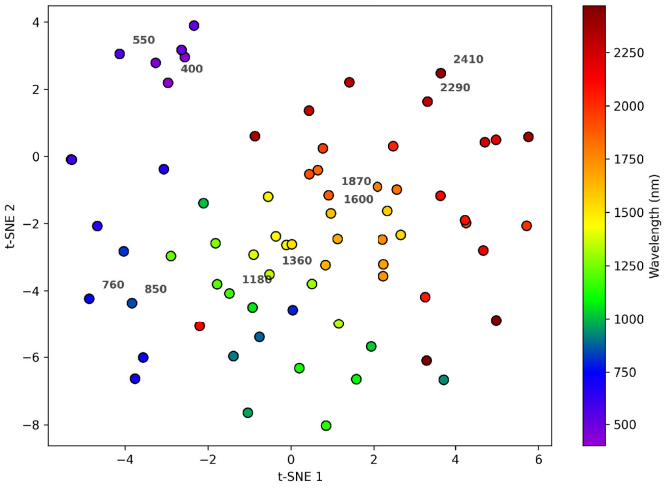


Fig. 7. Pretrained weights of wavelength-based spectral embedding: the high-dimensional layer weights of each GLP are converted into a 2-D space using t-SNE. The embedding weights of adjacent bands are distributed closely in the feature space.

smoothed to remove high-frequency noise by using a Savitzky-Golay filter [58] with a window length of 5 and a polyorder of 2.

2) Implementation Details: For the pretraining task, we use masked spectral modeling, adapted from the masked autoencoder (MAE) [59], which is widely used for pretraining spectral data. In this pretraining task, the input spectra are randomly masked, and the model is trained to reconstruct the missing reflectance values. Specifically, the input spectral data is first processed by the spectral harmonizer, which converts the unevenly distributed EnMAP bands into 210 uniformly spaced bands spanning 400–2500 nm with a 10-nm bandwidth. These 210 bands are divided into groups of three, and the masking operation is conducted randomly at the group level with an 80

3) Pretraining Results: Before evaluating the pretrained model on downstream datasets, we first assess the pretrained model's effectiveness by visualizing and comparing the reconstructed spectra and the original ones. As shown in Fig. 6, the reconstructed spectra align closely with the original ones, both in overall shape and fine spectral details. Such strong alignment indicates that the pretrained model can effectively capture spectral features and represent spectral information from hyperspectral pixels. Besides the reconstructed spectra, the pretrained weights of the wavelength-based embedding

TABLE I

SUMMARY OF FINETUNING DATASETS. FOR EACH DATASET, THE SENSOR TYPE, SPECTRAL RANGE, BAND NUMBER, SPATIAL RESOLUTION, IMAGE SIZE, AND TASKS ARE LISTED. FOR THE BAVARIA (SOC) DATASET, THE IMAGES VARY IN SIZE. THE GLORIA AND KARLY DATASETS ONLY CONTAIN HYPERSPECTRAL PIXELS

Finetuning Datasets	Sensor	Spectral Range (nm)	Band Number	Spatial Resolution (m)	Image Size	Task
Pavia University [41]	ROSIS	430 – 860	103	1.3	610 * 340	Land Cover Classification
Indian Pines [41]	AVIRIS	400 – 2500	224	20	145 * 145	Land Cover Classification
WHU-Hi-HanChuan [6]	Headwall Nano-Hyperspec	400 – 1000	274	0.109	1217 * 303	Land Cover Classification
WHU-Hi-HongHu [6]	Headwall Nano-Hyperspec	400 – 1000	270	0.043	940 * 475	Land Cover Classification
Hermison [75]	EO-1 Hyperion	400 – 2500	242	30	390 * 220	Change Detection
Bay Area [75]	AVIRIS	400 – 2500	224	16.9	600 * 500	Change Detection
Santa Barbara [75]	AVIRIS	400 – 2500	224	15.2	984 * 740	Change Detection
Augsburg [76]	HySpex	400 – 2500	180	30	332 * 485	Modality Fusion
Berlin [76]	HyMap	400 – 2500	244	30	797 * 220	Modality Fusion
Bavaria [42]	DESIS	400 – 1000	157	30	–	Soil Organic Carbon Regression
GLORIA [61]	Multiple Sensors	350 – 900	551	–	–	Water Chlorophyll Regression
KarLy [62]	Cubert UHD 285	450 – 950	125	–	50 * 50	Soil Moisture Regression

layers are also visualized in Fig. 7. For each GLP, the high-dimensional layer weights are transformed into a 2-D space using the t-SNE algorithm [60]. The figure shows the distribution of the embedding weights over the spectral range from 400 to 2500 nm. It can be observed that the embedding weights of spectral adjacent bands are distributed closely in the feature space. Such proximity suggests that the pretrained model can effectively capture the intrinsic spectral continuity of hyperspectral data, demonstrating the success of pretraining.

B. Finetuning

In the finetuning stage, we evaluate the pretrained models on various hyperspectral datasets acquired from different hyperspectral sensors. First, following the finetuning pipelines in Fig. 2(a), we conduct patch-based experiments on nine benchmark datasets over three different downstream applications, including land cover classification, temporal change detection, and modality fusion with SAR. Second, following the finetuning pipelines in Fig. 2(b), we assess the practical utility of proposed model on pixel-based regression tasks, including soil organic carbon (SOC) estimation over Bavaria [42], water chlorophyll estimation using the GLORIA dataset [61], soil moisture estimation on KarLy dataset [62]. All tasks are typical and challenging spectroscopy-based hyperspectral applications, where pixel-level spectra serve as inputs and pixel-wise annotations serve as labels. For all these finetuning experiments, the proposed FM is evaluated under different tuning settings, including training from scratch, linear probing, freezing the wavelength-based embedding layers, and fully fine-tuning. Moreover, the pretrained model is further compared with representative hyperspectral frameworks (3-D CNN [63] and SpectralFormer [38]), dedicated hyperspectral algorithms for specific downstream applications (classification: GAHT [64], SSFTT [65], and CLOLN [66]; regression: 1-D CNN [67]; change detection: HYGSTAN [68], GLAFormer [69], and SpectralKAN [70]; modality fusion: AsyFFNet [71], Fusion HCT [72], and MACN [73]), and hyperspectral FMs (HyperSL [26], HyperSigma [24], and SpectralEarth [25]). Further details are provided below:

1) Dataset: The fine-tuning datasets are collected from different sensors with varying spectral and spatial characteristics. To gain a clear overview of these datasets, we summarize the sensor type, spectral range, band number, spatial resolution, image size, and tasks in Table I. It is evident that there are substantial variations in sensor characteristics [74], highlighting the challenges and the necessity of developing sensor-agnostic hyperspectral FM.

For the patch-based datasets, nine widely used single-scene benchmark datasets are adopted: Pavia University, Indian Pines [41], WHU-Hi-HanChuan, and WHU-Hi-HongHu [6] for land cover classification, Hermiston, Bay Area, and Santa Barbara datasets [75] for change detection, Augsburg and Berlin datasets [76] for modality fusion with SAR. Following common practice in other hyperspectral studies, a limited number of labeled samples per class are randomly selected for training and validation: Specifically, for the four hyperspectral classification datasets, 20/20/50/50 samples per class are used for training, and 10/10/20/20 samples per class are used for validation, respectively. For the change detection task, 100 samples are selected from each class (changed and unchanged) for training, with an additional 50 samples per class for validation. For the two hyperspectral-SAR data fusion datasets, 50/50 samples per class are used for training, and 20/20 for validation. All remaining samples are used for testing in all experiments above.

For the pixel-based tasks, the FM is evaluated on the estimation of SOC, retrieval of water chlorophyll concentration, and soil moisture regression. The Bavaria-SOC dataset is derived from 850 DESIS hyperspectral images, which are distributed across the Bavaria region in Germany (~70 000 km^2). For each soil sample, the input spectrum is generated by building the hyperspectral temporal composite. The dataset is geographically split into 612 calibration samples (including both the training and validation sets) and 216 test samples. The GLORIA dataset is a globally representative hyperspectral in situ dataset for optical sensing of water quality. In total, it includes 7572 curated hyperspectral remote sensing reflectance measurements at 1-nm intervals from 350 to 900 nm. For finetuning, we select reflectance measurements with co-located chlorophyll a measurements, filtering values to the range of [0, 100] mg m^{-3}. This results in 871 samples: 697 for training and validation, and 174 for testing. The KarLy soil moisture dataset was collected in May 2017 in Karlsruhe, Germany. A total of 679 bare soil samples were collected, and soil moisture was measured as a percentage. The corresponding hyperspectral image was collected from a hyperspectral camera installed

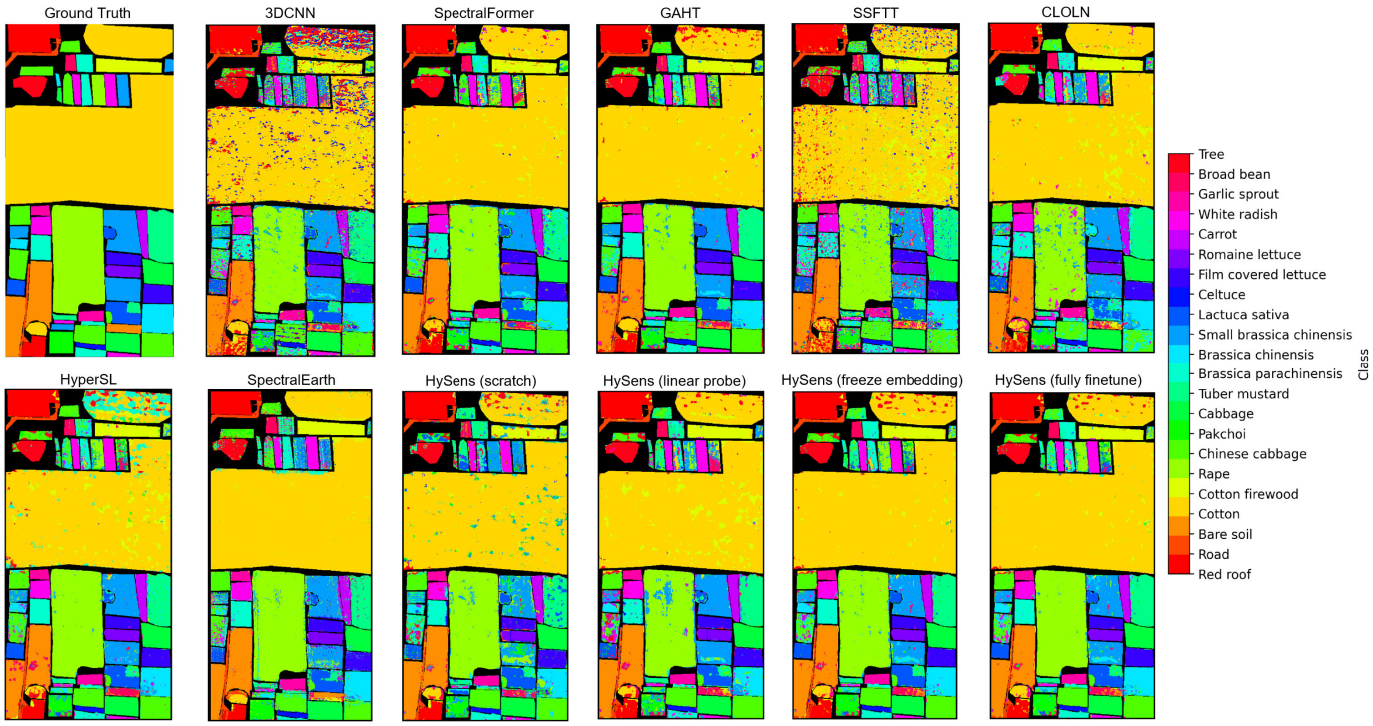


Fig. 8. Classification maps on WHU-Hi-HongHu: each classification map is generated using the best-performing model.

on a 1.70-m height tripod. Based on each image, the mean spectrum was computed and calibrated with the Spectralon spectrum.

2) Implementation Details: For all patch-based experiments, a patch size of 8×8 is used across all datasets. Especially, for the change detection tasks, features are first extracted separately from the prechange and postchange patches, and the feature difference is subsequently fed into the prediction head. For hyperspectra-SAR modality fusion experiments, SAR features are extracted using the same lightweight encoder comprised of two convolutional layers and are then concatenated with the corresponding hyperspectral features. These settings are applied to all compared methods except for task-specific dedicated models, where the original architectures are followed. Following other hyperspectral studies, the evaluation metrics used in hyperspectral classification and modality fusion include overall accuracy (OA) and average accuracy (AA). For change detection, OA, Kappa coefficient, precision, recall, and $F1$ score are computed. The proposed HySens FM is trained for up to 500 epochs with early stopping based on validation performance to prevent overfitting. Optimization is performed using the Adam optimizer with a learning rate of 2×10^{-5} and a weight decay of 0.05. For all baseline comparison methods, we maintain the same optimization settings as those in the original papers. Cross-entropy loss is adopted as the loss function. The best-performing models are all selected based on the highest OA on the validation set and will be further used to report the final test results. To ensure reliable evaluation, results are averaged over five runs. The mean results are reported.

For the pixel-based spectroscopy experiments, all inputs are 1-D pixel-wise hyperspectral spectra. To evaluate regression performance, metrics such as root-mean-square error (RMSE) and R^2 are used. The proposed FM is trained for up to 500 epochs with early stopping. We employ the Adam optimizer with a learning rate of 2×10^{-5} and a weight decay of 0.05. The mean squared error (mse) loss is used to optimize the regression models. The best-performing model is selected based on the lowest validation RMSE and then applied to get the test results. In particular, pixel-based spectroscopy tasks are more application-oriented rather than benchmark-driven, and there is a quite limited number of dedicated models for it. Therefore, in addition to comparing with other hyperspectral FMs, two networks are adopted as simple baseline methods: 1-D CNN model (CNN-based) [67] and SpectralFormer (transformer-based) model [38]. Regarding all the comparison methods, we follow their original optimization settings as closely as possible to ensure a fair and consistent comparison. It is noteworthy that pixel-based experiments are only conducted in models that support pixel-wise inputs. For the HyperSigma model, only the pretrained spectral encoder is used for pixel-wise tasks. To reduce the impact of randomness, the mean values of the results are reported from five independent runs for each model and setting.

3) Patch-Based Experimental Results: The experimental results for classification, change detection, and modality fusion are presented in Tables II–VI. Across all these datasets from different sensors, the proposed hyperspectral FM with frozen embedding layers and fully fine-tuning settings achieves the best performance compared to other baseline models and various fine-tuning strategies. Notably, the proposed FM with pretrained weights significantly outperforms the same framework trained from scratch, demonstrating the benefit of pretraining in extracting general spectral features.

TABLE II

PERFORMANCE ON PATCH-BASED HYPERSPECTRAL CLASSIFICATION DATASETS: THIS TABLE SHOWS THE CLASSIFICATION RESULTS ON DIFFERENT DATASETS USING DIFFERENT MODELS. THE AVERAGED OA AND AA ARE COMPUTED ON THE TEST SET AND PRESENTED IN THE TABLE. FOR EACH DATASET, THE BEST AND SECOND-BEST PERFORMANCES ARE HIGHLIGHTED

Dataset	Pavia U	Indian Pines	WHU-Hi-HanChuan	WHU-Hi-HongHu
Evaluation Metrics	**OA(%) / AA(%)**	**OA(%) / AA(%)**	**OA(%) / AA(%)**	**OA(%) / AA(%)**
3DCNN [63]	77.62 / 83.70	63.49 / 77.96	79.88 / 76.34	78.98 / 77.82
SpectralFormer [38]	76.99 / 81.67	71.67 / 82.72	84.39 / 83.07	89.09 / 87.16
GAHT [64]	89.17 / 90.15	77.76 / 88.66	88.80 / 86.98	91.28 / 90.66
SSFTT [65]	85.07 / 85.42	74.82 / 85.85	76.78 / 74.48	78.72 / 81.50
CLOLN [66]	88.11 / 92.27	80.23 / 87.45	80.99 / 78.19	85.43 / 84.60
HyperSL [26]	90.54 / 94.34	81.96 / 90.73	90.53 / 90.87	91.03 / 91.34
SpectralEarth [25]	85.37 / 87.82	81.34 / 88.64	91.15 / 90.50	87.13 / 85.97
HySens (scratch)	83.08 / 89.55	70.11 / 83.46	80.61 / 76.07	76.36 / 72.70
HySens (linear probe)	86.82 / 92.57	77.12 / 85.98	90.70 / 90.54	88.87 / 86.55
HySens (freeze embedding)	91.58 / 94.06	84.04 / 90.61	91.75 / 91.52	92.80 / 92.51
HySens (fully finetune)	90.45 / 93.71	82.48 / 89.18	92.35 / 92.56	92.48 / 92.28

TABLE III

PERFORMANCE ON PATCH-BASED HYPERSPECTRAL CHANGE DETECTION ON HERMISTON DATASET: THIS TABLE SHOWS THE CHANGE DETECTION PERFORMANCES USING DIFFERENT MODELS. THE AVERAGED OA, KAPPA, PRECISION, RECALL, AND $F1$ ARE COMPUTED ON THE TEST SET AND PRESENTED IN THE TABLE. THE BEST AND SECOND-BEST PERFORMANCES ARE HIGHLIGHTED

Evaluation Metrics	OA(%)	Kappa	Precision(%)	Recall(%)	F1(%)
3DCNN [63]	96.42	0.93	96.34	96.42	96.36
SpectralFormer [38]	96.49	92.85	96.44	96.43	96.42
HyGSTAN [68]	97.74	95.39	97.76	97.68	97.69
GLAFormer [69]	98.80	97.56	98.74	98.82	98.78
SpectralKAN [70]	95.16	90.17	95.01	95.18	95.08
HyperSL [26]	97.15	94.21	97.06	97.17	97.11
HyperSIGMA [24]	96.10	92.08	64.11	64.01	64.05
SpectralEarth [25]	90.72	81.25	90.81	91.00	90.58
HySens (scratch)	97.73	95.36	97.75	97.66	97.68
HySens (linear probe)	98.02	95.96	98.08	97.94	97.98
HySens (freeze embedding)	97.88	95.66	97.97	97.78	97.83
HySens (fully finetune)	98.23	96.39	98.27	98.15	98.20

Compared with other hyperspectral FMs, our proposed FM also shows better performance, highlighting the effectiveness of the designed wavelength-aware modules. These modules enable better transfer learning across datasets from different sensors. Especially when the pretrained wavelength-based embedding layers are frozen, the proposed FM shows excellent performance. Such an observation demonstrates the generality of the pretrained wavelength-based embedding layers, which can effectively capture wavelength-based spectral representations. When the fine-tuning datasets are small and limited, freezing the embedding layers reduces the number of trainable parameters compared to fully fine-tuning, resulting in slightly better performance on some datasets due to faster convergence on the training set. Moreover, dedicated models such as 3DCNN, GAHT, SSFTT, CLOLN, HyGSTAN, GALFormer, SpectralKAN, AsyFFNet, Fusion HCT, and MACN are all spectral–spatial hyperspectral models. Although our proposed model focuses primarily on spectral features without learning high-level spatial features, it achieves comparable or even higher performance across different downstream applications. This result demonstrates that the proposed model effectively extracts discriminative spectral information from hyperspectral data. By incorporating the spectral–spatial aggregator, the spectral–spatial feature can also be obtained by enriching the spectral features with local spatial information. Beyond the evaluation metrics, the classification maps on the WHU-Hi-HongHu dataset are visualized in Fig. 8. Each map is generated using the best model, and the inference is applied to the entire image. Compared to the ground-truth map, the proposed FM with frozen embedding weights and fully fine-tuned weights still shows the best alignment, further verifying its success in classification tasks. Both metrics and visualizations show the superior performance of the proposed FM across different sensors. These results highlight the effectiveness of both the framework design and the pretraining strategy for sensor-agnostic hyperspectral FM.

TABLE IV

PERFORMANCE ON PATCH-BASED HYPERSPECTRAL CHANGE DETECTION ON BAY AREA DATASET: THIS TABLE SHOWS THE CHANGE DETECTION PERFORMANCES USING DIFFERENT MODELS. THE AVERAGED OA KAPPA, PRECISION, RECALL, AND $F1$ ARE COMPUTED ON THE TEST SET AND PRESENTED IN THE TABLE. THE BEST AND SECOND-BEST PERFORMANCES ARE HIGHLIGHTED

Evaluation Metrics	OA(%)	Kappa	Precision(%)	Recall(%)	F1(%)
3DCNN [63]	72.91	45.62	72.82	72.82	72.81
SpectralFormer [38]	92.44	84.81	92.46	92.37	92.40
HyGSTAN [68]	94.44	88.88	94.59	94.52	94.43
GLAFormer [69]	96.43	92.84	96.40	96.48	96.42
SpectralKAN [70]	97.09	94.16	97.04	97.14	97.08
HyperSL [26]	90.48	80.87	90.49	90.42	90.43
HyperSIGMA [24]	84.85	69.86	56.89	56.61	56.72
SpectralEarth [25]	94.22	88.43	94.26	94.22	94.21
HySens (scratch)	96.21	92.41	96.27	96.29	96.20
HySens (linear probe)	96.85	93.69	96.90	96.82	96.84
HySens (freeze embedding)	96.80	93.56	96.93	96.72	96.78
HySens (fully finetune)	96.79	93.55	96.93	96.71	96.77

TABLE V

PERFORMANCE ON PATCH-BASED HYPERSPECTRAL CHANGE DETECTION ON SANTA BARBARA DATASET: THIS TABLE SHOWS THE CHANGE DETECTION PERFORMANCES USING DIFFERENT MODELS. THE AVERAGED OA, KAPPA, PRECISION, RECALL, AND $F1$ ARE COMPUTED ON THE TEST SET AND PRESENTED IN THE TABLE. THE BEST AND SECOND-BEST PERFORMANCES ARE HIGHLIGHTED

Evaluation Metrics	OA(%)	Kappa	Precision(%)	Recall(%)	F1(%)
3DCNN [63]	80.52	59.65	79.57	80.22	79.80
SpectralFormer [38]	90.57	80.58	89.90	91.03	90.27
HyGSTAN [68]	95.72	91.01	95.51	95.52	95.51
GLAFormer [69]	95.80	91.27	95.38	96.03	95.63
SpectralKAN [70]	96.58	92.82	96.48	96.36	96.41
HyperSL [26]	91.34	82.36	91.08	91.99	91.12
HyperSIGMA [24]	90.04	79.54	60.46	59.94	60.16
SpectralEarth [25]	92.78	84.93	91.99	93.16	92.46
HySens (scratch)	97.04	93.74	97.27	96.56	96.87
HySens (linear probe)	97.33	94.36	97.39	97.01	97.18
HySens (freeze embedding)	97.41	94.53	97.46	97.10	97.27
HySens (fully finetune)	97.43	94.58	97.46	97.14	97.29

4) Pixel-Based Experimental Results: For the pixel-based spectroscopy tasks, the experimental results are summarized in Table VII. In the challenging task of estimating SOC, FMs like HyperSigma, HyperSL, and SpectralEarth struggle to fit the data, resulting in very low R^2 scores. In contrast, our proposed FM with fully finetuning achieves the best performance with RMSE 1.06 and R^2 0.39. In particular, in the article [42], reasonable results are only obtained after complex feature engineering, whereas the pretrained FM can directly use the original hyperspectral reflectance, substantially simplifying the entire process for downstream users. The success of the proposed FM can be attributed to its sensor-agnostic design, which enables effective adaptation across heterogeneous hyperspectral datasets. Specifically, for the Bavaria-SOC dataset, the hyperspectral data are acquired from the DESIS satellite, whose sensor characteristics differ significantly from those of the pretrained datasets used by other hyperspectral FMs and our model. Extracting fine-grained spectral features is also necessary for accurate SOC regression, which necessitates considering sensor discrepancies when transferring the pretrained model. The proposed FM addresses this challenge by leveraging wavelength-aware embedding layers that explicitly account for sensor differences and enhance model transferability. Furthermore, our proposed model outperforms the 1-D CNN and the SpectralFormer model, both trained from scratch, highlighting the benefits of pretraining for small-scale datasets—a common scenario in real-world hyperspectral applications. For the test results on the GLORIA dataset for water chlorophyll estimation, since the dataset is well curated and filtered, all models achieve reasonable performance. Among them, the proposed FM with fully finetuned weights or frozen embeddings achieves the best performance,

TABLE VI

PERFORMANCE ON PATCH-BASED HYPERSPECTRAL-SAR MODALITY FUSION DATASETS: THIS TABLE SHOWS THE CLASSIFICATION RESULTS ON DIFFERENT MODALITY FUSION DATASETS USING DIFFERENT MODELS. THE AVERAGED OA AND AA ARE COMPUTED ON THE TEST SET AND PRESENTED IN THE TABLE. FOR EACH DATASET, THE BEST AND SECOND-BEST PERFORMANCES ARE HIGHLIGHTED

Dataset	Augsburg	Berlin
Evaluation Metrics	OA(%) / AA(%)	OA(%) / AA(%)
3DCNN [63]	94.10 / 95.45	94.70 / 95.22
SpectralFormer [38]	92.65 / 94.14	93.19 / 94.21
AsyFFNet [71]	96.15 / 97.01	96.60 / 96.72
Fusion HCT [72]	97.15 / 97.66	96.65 / 96.63
MACN [73]	95.45 / 96.60	96.60 / 96.80
HyperSL [26]	88.51 / 81.58	72.06 / 76.88
HyperSIGMA [24]	90.26 / 66.39	71.46 / 58.55
SpectralEarth [25]	90.76 / 93.64	97.55 / 98.06
HySens (scratch)	93.60 / 95.06	96.58 / 96.92
HySens (linear probe)	97.30 / 97.76	98.18 / 98.41
HySens (freeze embedding)	97.75 / 98.32	98.79 / 98.91
HySens (fully finetune)	97.80 / 98.36	98.88 / 99.04

demonstrating the generality of the pretrained spectral representation for pixel-wise hyperspectral tasks across diverse spectroscopy applications. For the regression results on the KarLy dataset of soil moisture retrieval, the pretrained FM with fully finetuned weights shows the best performance, benefiting from its ability to extract spectral features across sensors, which are fine-grained and relevant for soil moisture. Overall, these pixel-based experimental results demonstrate the practical applicability of the proposed FM across a wide range of challenging downstream hyperspectral applications, further confirming the effectiveness of the pretrained, sensor-agnostic hyperspectral FM.

C. Ablation Study

In addition to the above fine-tuning tasks, ablation studies are conducted to investigate the effectiveness of the wavelength-based embedding layers and to assess the interpretability and overall efficiency of the proposed framework.

1) Wavelength-Based Modules: In the proposed framework, the wavelength-based embedding layers are designed to be wavelength-aware and could be easily transferred across various sensors based on the matched band wavelengths. To further verify that these embedding layers are truly wavelength-aware and that transfer learning indeed benefits from the aligned wavelengths, we perform an ablation study in which the pretrained embedding layers are transferred under deliberately mismatched band wavelengths. Specifically, we shift the aligned embedding layers by different numbers of band groups (10, 20, and 30), creating experimental conditions for transfer learning across mismatched band wavelengths. We then compare the finetuning performance with results obtained with correctly matched wavelengths. The experimental results on the Pavia University dataset are shown in Fig. 9. In the fine-tuning results of linear probing, freeze embedding, and fully fine-tuning, a clear performance drop is observed when the pretrained embedding layers are transferred to mismatched bands. Moreover, the degradation increases in magnitude as

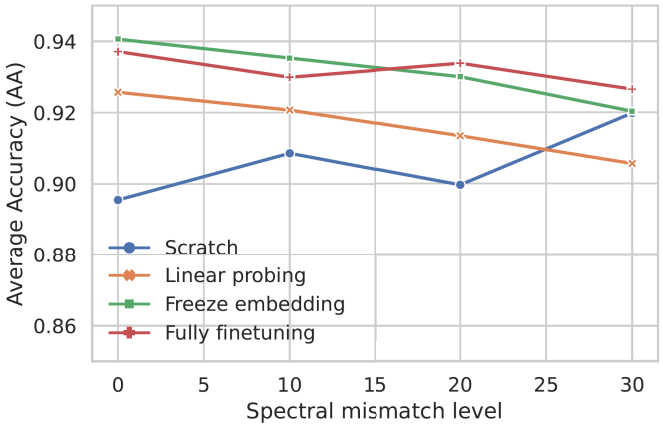


Fig. 9. Classification performances under mismatched embedding layers: the ablation experiments are conducted on the Pavia University dataset with AA as the evaluation metric. By shifting the aligned wavelength-based embedding layers over different numbers of band groups, we create different mismatch levels, such as 10, 20, and 30. The experiments are conducted over all finetuning settings.

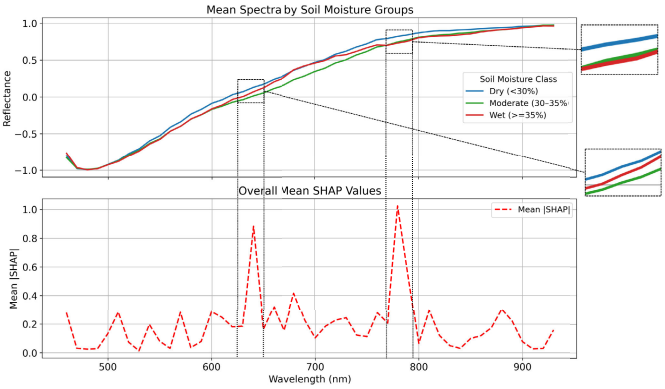


Fig. 10. SHAP values across spectral bands for soil moisture estimation: The spectra of the test set are classified into three groups based on soil moisture values, and the averaged spectra are visualized. The absolute SHAP values are computed using the fully finetuned FM across spectral bands.

the degree of mismatch increases. For training from scratch, performance fluctuates without a decrease, as the framework's structures remain unchanged despite mismatched levels. These findings suggest that the proposed embedding layers effectively capture wavelength-aware representations and can be transferred across different sensors when the band wavelengths are correctly aligned.

2) Band Importance for Spectroscopy Tasks: From the results presented in Section IV-B4, the proposed sensor-agnostic FM exhibits superior performance, particularly in pixel-wise spectroscopy regression tasks. To further assess the explainability of the pretrained FM, the band importance is analyzed for both soil moisture and water chlorophyll estimation, as illustrated in Figs. 10 and 11, respectively. Specifically, absolute SHAP values [77] are calculated using the fully finetuned models and test spectra from the KarLy and Gloria datasets. For soil moisture estimation in Fig. 10, spectral bands in the range of 640–650 nm and 770–780 nm show the highest SHAP values, indicating their strong contribution to model prediction. Such findings align with the discriminative spectral features observed in the spectra figure, where wet

TABLE VII

PERFORMANCE ON PIXEL-BASED HYPERSPECTRAL DATASETS: THIS TABLE SHOWS THE RESULTS ON SPECTROSCOPY-BASED TASKS, INCLUDING SOC ESTIMATION, WATER CHLOROPHYLL ESTIMATION, AND SOIL MOISTURE RETRIEVAL. $RMSE$ AND $/R^2$ ARE SELECTED AS EVALUATION METRICS FOR REGRESSION. THE PRESENTED RESULTS ARE THE TEST PERFORMANCE USING THE BEST MODEL FROM THE VALIDATION SET. FOR EACH DATASET, THE BEST AND SECOND-BEST PERFORMANCES ARE HIGHLIGHTED

Dataset	**Bavaria (Soil Organic Carbon)**	**Gloria (Water Chlorophyll)**	**KarLy (Soil Moisture)**
Evaluation Metrics	$RMSE$ / R^2	$RMSE$ / R^2	$RMSE$ / R^2
1D CNN [67]	1.07 / 0.38	9.65 / 0.82	3.54 / 0.12
SpectralFormer [38]	1.26 / 0.13	10.36 / 0.79	2.48 / 0.57
HyperSL [26]	1.79 / -0.77	9.85 / 0.81	3.40 / 0.16
HyperSigma [24]	1.39 / -0.08	9.61 / 0.82	1.83 / 0.76
SpectralEarth [25]	2.27 / -1.89	10.21 / 0.80	6.67 / -2.15
HySens (scratch)	1.17 / 0.25	8.73 / 0.85	2.40 / 0.60
HySens (linear probe)	1.25 / 0.16	9.60 / 0.82	3.23 / 0.27
HySens (freeze embedding)	1.08 / 0.36	8.76 / 0.85	1.70 / 0.80
HySens (fully finetune)	1.06 / 0.39	8.53 / 0.86	1.69 / 0.80

TABLE VIII

MODEL EFFICIENCY COMPARISON

Hyperspectral FMs	**Number of Parameters (M)**	**Inference Latency (ms / batch)**	**Peak GPU Memory (MB)**	**Finetuning Time (s)**
HyperSL (small) [26]	11.05	164.44	2047.9	2873.36
HyperSigma (base) [24]	182.76	47.16	883.0	993.38
SpectralEarth (Spec.RN50) [25]	24.22	7.01	251.3	505.17
HySens	7.52	23.09	100.5	712.05

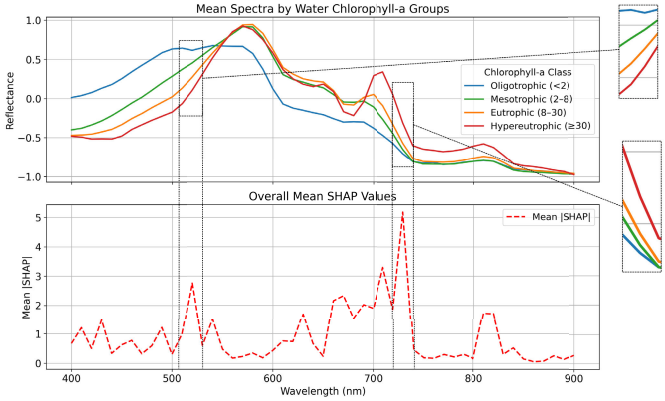


Fig. 11. SHAP values across spectral bands for water chlorophyll estimation: The water chlorophyll values are grouped into four classes. The resulting mean spectra are visualized in the figure, while the mean absolute SHAP values are generated using the fully finetuned FM.

soils display a steeper reflectance increase in the 640–650-nm range and a more saturated NIR plateau at 770–780 nm. For water chlorophyll estimation in Fig. 11, the spectral bands of 720–730 nm and 520–530 nm have high SHAP values, where the mean spectra across different chlorophyll levels show distinctly varying decreasing and increasing slopes. The correspondence between SHAP-derived band importance and the discriminative spectral features confirms the pretrained model's strong explainability and demonstrates its ability to capture the underlying physically aligned spectral features for spectroscopy tasks.

shown in the figure, and the mean absolute SHAP values are computed

3) Model Efficiency: For FMs, model efficiency is important when applied to various downstream tasks. In particular, many hyperspectral downstream datasets are small and have limited labeled samples. In such scenarios, the domain users would expect the FM not only to achieve strong performance but also to be lightweight and easy to fine-tune. To assess the efficiency of our proposed FM, the number of parameters, inference latency, peak GPU memory, and fine-tuning time are compared with other hyperspectral FMs. To have a fair comparison, all models are trained on a single NVIDIA A40 GPU with FP32 precision, a batch size of 16, an epoch of 500, and a learning rate of 2×10^{-5}. The results are presented in Table VIII. The proposed HySens model demonstrates strong overall efficiency, having the best parameter and GPU memory efficiency while remaining competitive in inference latency and finetuning time. Such comparison results show that the proposed HySens model is efficient to use, which strengthens the practical applicability for domain users with limited computation resources.

V. CONCLUSION

In this study, a generic sensor-agnostic FM for hyperspectral data is successfully developed. Specifically, 1) the introduced spectral harmonizer align various band distributions onto a unified spectral grid and reduce the spectral domain shifts while transferring pretrained models; 2) the wavelength-based spectral embeddings make each band group customized embedded, and such embedding layers could be transferred based on the band wavelengths of various sensors; and 3) the model extracts the spectral features effectively by treating the hyperspectral spectra as sequences and using a transformer-based spectral encoder. The developed HySens FM shows superior performance compared with SOTA hyperspectral algorithms and hyperspectral FMs across different tasks and sensors. Moreover, the proposed model is easily

transferable, having good interpretability and efficiency. Overall, the developed sensor-agnostic FM demonstrates strong cross-sensor transferability and high computational efficiency, making it a promising FM for practical deployment in real-world hyperspectral applications.

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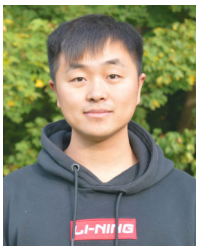
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